

RESEARCH ARTICLE

A Hybrid Transformer-Based Framework for Multi-Document Summarization of Turkish Legal Documents

MAHA AHMED ABDULLAH ALBAYATI^{ID} AND OGUZ FINDIK^{ID}

Department of Computer Engineering, Karabük University, 78000 Karabük, Türkiye

Corresponding author: Maha Ahmed Abdullah Albayati (mahaa6022@gmail.com)

ABSTRACT The rapid growth in the volume and complexity of legal documents has created an urgent need for automated tools to help legal professionals efficiently extract and summarize critical information. While numerous studies have addressed this issue across various languages, Turkish legal document summarization presents unique challenges due to the language's agglutinative structure and domain-specific terminology. This gap is particularly significant, given Turkey's extensive legal system and the increasing volume of digital legal documents. This study represents the first comprehensive effort to address this gap by introducing a hybrid approach that combines traditional extractive techniques, including TF-IDF and TextRank, with advanced transformer-based models such as LED, Long-T5, BART-large, and GPT-3.5 Turbo. A new dataset of 2,000 Turkish civil cases, carefully curated and validated from publicly accessible legal platforms, was developed to support this research. Extractive summarization methods were evaluated based on cosine similarity, content precision, recall, and F1-score, with TF-IDF achieving a content F1-score of 0.47 using keyword-based summaries. Transformer-based models demonstrated significant improvements, with GPT-3.5 Turbo achieving the highest ROUGE scores (ROUGE-1: 55%, ROUGE-2: 35%, ROUGE-L: 42%, ROUGE-Sum: 44%), showcasing its superior ability to generate accurate and coherent summaries. By tackling the unique linguistic challenges of Turkish agglutinative morphology, this research provides the first foundation for scalable, automated tools to enhance legal professionals' efficiency and decision-making within Turkey's legal system.

INDEX TERMS Extractive and abstractive summarization, legal document processing, multi-document summarization, transformer models, Turkish legal texts.

I. INTRODUCTION

Natural Language Processing (NLP) has been recognized as an essential area of artificial intelligence, focusing on enabling machines to understand, interpret, and generate human language [1]. A wide range of tasks, such as sentiment analysis [2], machine translation, Named Entity Recognition (NER), and text summarization, are encompassed within NLP. As textual data grows exponentially in fields like law, medicine, and business, an increasing demand for efficient methods to process and extract useful information from

large text corpora has been noted [3], [4], [5]. One of the key applications of NLP, text summarization, is aimed at condensing a large body of text into a shorter version while retaining its essential meaning. Text summarization is typically divided into two main categories: single-document and multi-document summarization [6], [7]. In single-document summarization, a concise summary is generated from a single source of information, such as an article or legal case [6]. This approach is well-suited for situations where the source document is relatively self-contained and limited in scope. Multi-document summarization, on the other hand, involves the creation of a summary from multiple related documents, a far more complex task due to the

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need to integrate information from various sources while avoiding redundancy and ensuring coherence [8]. Multi-document summarization is particularly useful in legal and other professional fields, where decisions or findings may span multiple texts and documents. In the legal domain, the summarization of legal documents presents unique challenges due to their complexity, length, and specialized language [9]. Legal texts often contain intricate details, precise terminology, and specific legal reasoning that must be preserved in any summary. Three primary summarization techniques are used to tackle these challenges: extractive, abstractive, and hybrid methods.

Extractive summarization involves the selection of key sentences directly from the original document without modifying them [10], [11], [12]. Traditional machine learning techniques, such as TF-IDF (Term Frequency-Inverse Document Frequency) and graph-based models like TextRank, are commonly used in extractive methods. These approaches rank sentences based on their importance within the document, with the highest-ranking sentences included in the summary [12], [13], [14], [15], [16]. Additionally, machine learning models, like Support Vector Machines (SVM) [17], Naive Bayes classifiers [18], and Random Forests, have been employed to automatically identify the most relevant parts of legal texts by training on labeled datasets [19], [20]. These models extract essential information from word frequency, sentence position, and similarity. More advanced models, such as LSTM (Long Short-Term Memory) networks and Recurrent Neural Networks (RNNs), are employed to help capture the sequential nature of the text, improving the extraction of key information more contextually [21], [22]. While extractive methods are regarded as effective at retaining the core meaning of the text, they often result in summaries that lack coherence and fluency, mainly when applied to longer documents or multiple texts. NER models are usually integrated into extractive summarization processes to enhance the extraction quality. NER models, such as those based on BERT (Bidirectional Encoder Representations from Transformers), are trained to identify and extract specific entities (such as names of persons, organizations, dates, or legal references) from the text. Identifying named entities is particularly useful in legal documents, which is critical for accurate summarization. NER models help ensure that key legal entities and terms are preserved in summary, further improving the accuracy of extractive methods [23], [24].

Abstractive summarization, on the other hand, is designed to generate new sentences that paraphrase the original content [25]. This approach goes beyond mere sentence extraction by understanding the meaning of the text and producing a concise version in a more natural and human-like manner. Abstractive models are heavily reliant on transformer-based architectures, such as BART (Bidirectional and Auto-Regressive Transformers) [26], T5 (Text-To-Text Transfer Transformer) [27], and

PEGASUS [28], which are pre-trained on large datasets and fine-tuned for summarization tasks. These models use Attention mechanisms to capture long-range dependencies in the text, generating coherent summaries that rephrase the original content. For example, GPT (Generative Pre-trained Transformer) [29] has demonstrated impressive performance in generating fluent, human-like text. However, challenges have been identified in maintaining factual accuracy in legal contexts where precision is critical. Abstractive methods, while more flexible and capable of producing cohesive summaries, often struggle with the issue of “hallucination” [30], where the model generates information not present in the original text. Hallucination is particularly problematic in domains like legal summarization, where accuracy is critical. To mitigate this, machine learning techniques can be leveraged to enhance abstractive summarization models. Machine learning models can be trained better to understand the structure and content of legal documents and learn patterns in the data that help ensure that critical details—such as court names, legal statutes, and case numbers—are accurately represented in the summaries [31]. Machine learning models can improve paraphrasing while retaining essential legal information by training on large, domain-specific datasets. These models can be fine-tuned with various objectives to reduce hallucination, ensuring the generated summaries align with the source text. Advanced techniques, like reinforcement learning [32], attention mechanisms, and encoder-decoder frameworks, allow machine learning-based abstractive summarization models to generate more reliable and coherent summaries [33]. This approach minimizes inaccuracies while ensuring the content remains informative and relevant to the legal context.

Hybrid summarization techniques have been developed to overcome the limitations of both extractive and abstractive approaches, combining the strengths of both methods. Hybrid models use extractive techniques to identify a document’s most important sentences or sections and then employ abstract methods to rephrase or compress the extracted content into a more coherent and fluent summary. Recent advancements in hybrid summarization are observed in models like LED (Longformer Encoder-Decoder) and Long-T5, specifically designed to handle long documents, making them ideal for the lengthy and complex nature of legal texts. These models extend the self-attention mechanism to process significant texts, such as legal documents while ensuring that the summary remains precise and readable. Other hybrid models are integrated with reinforcement learning techniques to fine-tune the summary generation process, rewarding the model for producing coherent and informative summaries [34], [35].

Despite the advancements in these summarization techniques, applying them to non-English languages, like Turkish, remains a significant challenge. Turkish legal documents, with their rich morphology and agglutinative structure [36], require additional adaptation and fine-tuning

of existing models. Moreover, the lack of large, annotated datasets for Turkish legal texts has further complicated the development of accurate summarization systems. However, the use of transformer-based models is seen as offering a promising solution, as these models can be fine-tuned to handle the specific linguistic features of Turkish. Combining extractive and abstractive methods in hybrid models can improve summarization systems and preserve legal integrity better while generating readable summaries, even for complex, underresourced languages like Turkish. In summary, extractive summarization is recognized for excelling at retaining factual accuracy, abstractive summarization is known for producing more fluent and natural language, and hybrid models are noted for blending both strengths to generate summaries that are considered coherent and legally precise. NER models play a crucial role in these methods by ensuring that essential legal entities are identified and preserved in the final summaries. These advancements in machine learning, transformer-based approaches, and NER models represent a significant step forward in automating legal document summarization, particularly for non-English languages like Turkish. When examining the landscape of research on legal document summarization, a wealth of studies has been conducted for various languages, such as English, Dutch, Portuguese, Spanish, Korean, and Italian [37]. These studies have demonstrated varying degrees of success depending on the language and the availability of resources. For instance, legal document summarization in English has seen significant advancements due to the availability of large, annotated datasets and pre-trained models, leading to high accuracy and coherence in summaries. Research efforts in languages such as Dutch and Portuguese have produced promising results. However, the success of the models has often been limited by the complexity of legal texts and the lack of domain-specific datasets [38], [39]. Korean legal document summarization has also progressed, but models struggle to capture legal nuances, particularly in abstractive tasks, accurately [40]. In Spanish [41], while some models have been adapted to perform reasonably well, the availability of annotated data and pre-trained models remains a limiting factor, preventing the widespread success seen in English-language models.

However, despite these advancements in text summarization for other non-English languages, Turkish legal documents remain a largely unexplored area. No previous research has addressed the challenges of summarizing Turkish legal cases, particularly in multi-document summarization. This absence is significant because Turkish legal texts, unlike those in many other languages, present unique difficulties due to their agglutinative structure, complex legal terminology, and the limited availability of annotated datasets and pre-trained models. Existing summarization methods, such as TF-IDF and TextRank, have demonstrated effectiveness in other domains but have not been tailored to Turkish legal texts' structural and linguistic complexities. Similarly, while transformer-based

models have shown promise in summarizing legal documents in other languages, they require domain-specific adaptation to handle the nuances of Turkish legal texts effectively.

This study addresses these challenges by proposing a novel approach to multi-document summarization of Turkish legal cases, explicitly focusing on extracting court decisions—a critical aspect often overlooked by general methods. Our hybrid methodology combines TF-IDF- and TextRank-based extractive summarization to identify key legal sentences, while transformer-based abstractive models—including LED, Long-T5, BART-large, and GPT-3.5 Turbo—generate coherent and concise summaries. This dual approach bridges the gap between purely extractive and abstractive methods, leveraging the precision of extractive techniques to capture essential legal elements and the fluency of abstractive models to generate readable and comprehensive summaries. Among the transformer-based models evaluated, GPT-3.5 Turbo demonstrated the highest performance, achieving superior accuracy and fluency in generating legal summaries. This research represents the first attempt to systematically address the summarization of Turkish legal documents, offering a groundbreaking solution to a previously unexplored problem. By adapting and combining existing methods to meet the specific challenges of Turkish legal texts, our work provides a robust solution for the Turkish legal system. It establishes a framework for future advancements in automating legal document analysis for underrepresented languages.

The key contributions of this work include (1) the first comprehensive Turkish legal document summarization framework, (2) a validated dataset of Turkish civil cases, and (3) empirical evidence supporting the effectiveness of hybrid approaches for legal document summarization. This research establishes a foundation for developing practical tools to enhance legal professionals' efficiency and decision-making capabilities within Turkey's legal system.

The rest of this paper is organized as follows: Section II presents the materials and methods, detailing dataset collection, preprocessing steps, and the methodologies employed for extractive and abstractive summarization. It also outlines the evaluation metrics used to assess summarization performance. Section III discusses the results and provides a comprehensive analysis, including ROUGE metric performance, model comparisons, qualitative and quantitative evaluations of summaries, and the challenges encountered in multi-document summarization. It also addresses the limitations of the hybrid model and explores future research directions. Finally, Section IV concludes the paper by summarizing the key findings and emphasizing their significance in advancing legal document summarization.

II. MATERIAL AND METHOD

This section presents the methodological approach, including dataset preparation, preprocessing steps, and extractive and

abstract summarization techniques. The evaluation metrics used to assess the performance of the proposed framework are also described.

A. DATASET COLLECTION

The dataset used in this study comprises 2,000 Turkish legal documents retrieved from a publicly accessible website hosting various court rulings, including cases related to consumer rights (“tüketici hakları”). A Python-based web scraping tool was developed to automate the retrieval of these documents using specific keywords. Python’s requests library interacted with the website and fetched documents based on their unique document IDs. Extracted data was processed to isolate the relevant decision text, using BeautifulSoup for cleaning and formatting. As the dataset is publicly available and does not contain personal or sensitive information, no legal permissions or ethical approvals were required for its collection. Future work aims to expand the dataset to include a broader range of legal topics, such as criminal, administrative, and constitutional law, to enhance the model’s generalization and robustness. Additionally, efforts will focus on curating a more diverse and balanced dataset to address potential biases and ensure adequate representation of underrepresented legal scenarios.

B. PREPROCESSING

The preprocessing of Turkish legal documents involved several steps to optimize the text for analysis while preserving its linguistic and legal integrity. The Zemberek library [42], a specialized toolkit for processing Turkish texts, was employed for its robust support of morphological analysis, stemming, tokenization, and sentence extraction. Below is an outline of the preprocessing pipeline:

1) DOCUMENT CLEANING

The first step focused on removing irrelevant sections from the documents, such as headers, metadata, and procedural information, which are common in Turkish legal texts but do not contribute to the core legal arguments or court decisions. This process ensured that only the relevant legal content was retained for analysis.

2) TOKENIZATION

Following document cleaning, tokenization was performed using Zemberek’s TurkishTokenizer and TurkishSentence-Extractor tools. These tools divided the text into coherent sentences and further split it into individual tokens, facilitating manageable units for subsequent analysis.

3) STOP WORD REMOVAL

A custom list of Turkish stop words was applied to eliminate non-informative tokens. This list, loaded from an external file, included common words with minimal semantic weight in legal contexts. This step ensured the analysis focused solely on meaningful and legally relevant content.

4) NORMALIZATION

Several normalization steps were applied to standardize the text and address common formatting issues in legal documents:

- Numeric references, such as case numbers, dates, and monetary values, were removed to focus on the textual content of court decisions.
- Unnecessary punctuation marks (e.g., commas, periods, colons) and symbols were eliminated.
- Repeated words and phrases were consolidated into a single occurrence to avoid redundancy.
- Lowercasing: All text was converted to lowercase to ensure consistency.

5) MORPHOLOGICAL ANALYSIS AND STEMMING

Using Zemberek’s TurkishMorphology module, a morphological analysis was performed to identify each token’s root forms or lemmas. However, stemming was excluded from the preprocessing pipeline to preserve the legal integrity and accuracy of the text. In Turkish legal documents, inflected forms of words often carry significant semantic weight, which is critical for interpreting court decisions. For instance, verb tenses and case endings can drastically alter the meaning of legal statements. Thus, the decision to omit stemming was made to retain the full semantic context of the legal text.

6) FINAL DOCUMENT PREPARATION

After completing the cleaning, tokenization, stop word removal, normalization, and morphological analysis, the final version of the text was prepared for analysis. All irrelevant or unrelated content was stripped, leaving a clean, focused dataset suitable for summarization. The prepared dataset prioritized the court’s decisions and key legal arguments. An example is shown in Table 1.

C. EXTRACTIVE SUMMARIZATION

Extractive summarization involves selecting key sentences or sections directly from the source text to create a summary while preserving the original wording. This approach is particularly advantageous in the legal domain, where precision and maintaining the integrity of the original text are critical. This study employed extractive summarization to identify the most important sentences related to the court’s final rulings. Given legal texts’ repetitive and formulaic nature, this method effectively distilled pertinent information, including judgments, decrees, and critical legal arguments. Two primary techniques were utilized to perform extractive summarization on legal documents: Term Frequency-Inverse Document Frequency (TF-IDF) and TextRank. These techniques identify sentences containing frequently occurring words within the text, enabling the extraction of sentences that represent the document’s content. However, due to the generic nature of these methods, sentences not directly related to the court’s decisions may also be extracted. The summarization process was enhanced by incorporating

TABLE 1. Preprocessing steps for turkish legal document summarization.

Input:
13. Hukuk Dairesi 2015/38256 E. , 2017/11808 K. "İçtihat Metni" MAHKEMESİ :Asliye Hukuk Mahkemesi (Tüketici Mahkemesi Sıfatıyla) Taraflar arasındaki alacak davasının yapılan yargılaması sonunda ilamda yazılı nedenlerden dolayı davanın kabulüne yönelik olarak verilen hükmün süresi içinde davalı avukatınca temyiz edilmesi üzerine dosya incelendi gereği konuşulup düşünüldü. K A R A R Davacı, davalı ile 09.11.2017 tarihli EPT 1207 nolu devremülk satış sözleşmesi imzaladığını, toplamda 8.300,00 TL ödeme yaptığını, devre mülkün tesliminin yapılmadığını belirterek devremülk sözleşmesinin iptalini ve ödenen 8.300,00 TL'nin ödeme tarihinden itibaren yasal faizi ile birlikte iadesini talep etmiştir.
Sentences & Words Tokenizing:
['Taraflar arasındaki alacak davasının yapılan yargılaması sonunda ilamda yazılı nedenlerden dolayı davanın kabulüne yönelik olarak verilen hükmün süresi içinde davalı avukatınca temyiz edilmesi üzerine dosya incelendi gereği konuşulup düşünüldü.', 'K A R A R\\ndavacı, davalı ile 09.11.2017 tarihli EPT 1207 nolu devremülk satış sözleşmesi imzaladığını, toplamda 8.300,00 TL ödeme yaptığını, devre mülkün tesliminin yapılmadığını belirterek devremülk sözleşmesinin iptalini ve ödenen 8.300,00 TL'nin ödeme tarihinden itibaren yasal faizi ile birlikte iadesini talep etmiştir.']
['Taraflar', 'arasındaki', 'alacak', 'davasının', 'yapılan', 'yargılaması', 'sonunda', 'ilamda', 'yazılı', 'nedenlerden', 'dolayı', 'davanın', 'kabulüne', 'yönelik', 'olarak', 'verilen', 'hükmün', 'süresi', 'içinde', 'davalı', 'avukatınca', 'temyiz', 'edilmesi', 'üzerine', 'dosya', 'incelendi', 'gereği', 'konuşulup', 'düşünüldü', ' ', 'KARAR', 'Davacı', ' ', 'davalı', 'ile', '09.11.2017', 'tarihli', 'EPT', '1207', 'nolu', 'devremülk', 'satış', 'sözleşmesi', 'imzaladığını', ' ', 'toplamda', '8.300', ' ', '00', 'TL', 'ödeme', 'yaptığını', ' ', 'devre', 'mülkün', 'tesliminin', 'yapılmadığını', 'belirterek', 'devremülk', 'sözleşmesinin', 'iptalini', 've', 'ödenen', '8.300', ' ', '00', 'TL'nin', 'ödeme', 'tarihinden', 'itibaren', 'yasal', 'faizi', 'ile', 'birlikte', 'iadesini', 'talep', 'etmiştir', ' ']
After Removing Stop Word:
['taraflar', 'arasındaki', 'alacak', 'davasının', 'yargılaması', 'ilamda', 'yazılı', 'nedenlerden', 'davanın', 'kabulüne', 'yönelik', 'verilen', 'hükmün', 'süresi', 'içinde', 'davalı', 'avukatınca', 'temyiz', 'üzerine', 'dosya', 'incelendi', 'gereği', 'konuşulup', 'düşünüldü', ' ', 'karar', 'davacı', ' ', 'davalı', 'tarihli', 'ept', 'devremülk', 'satış', 'sözleşmesi', 'imzaladığını', ' ', 'toplamda', ' ', 'ödeme', ' ', 'devre', 'mülkün', 'tesliminin', 'yapılmadığını', 'belirterek', 'devremülk', 'sözleşmesinin', 'iptalini', 'ödenen', ' ', 'ödeme', 'tarihinden', 'yasal', 'faizi', 'iadesini', 'talep', 'etmiştir', ' ']
Normalization:
taraflar arasındaki alacak davasının yargılaması ilamda yazılı nedenlerden davanın kabulüne yönelik verilen hükmün süresi içinde davalı avukatınca temyiz üzerine dosya incelendi gereği konuşulup düşünüldü. karar davacı, davalı tarihli ept devremülk satış sözleşmesi imzaladığını, toplamda, ödeme, devre mülkün tesliminin yapılmadığını belirterek devremülk sözleşmesinin iptalini ödenen, ödeme tarihinden yasal faizi iadesini talep etmiştir.
Stemming:
taraf ara alacak davas yargıla ilam yazı neden dava kabul yönelik veri hükm süre iç dava avukat temyiz üzer dosya incel gereğ konuş düşün . karar dava , dava tarih ept devremülk sat söz imzala , toplam , öde , devre mülk teslim yap belir devremülk söz iptal öde , öde tarihi yasal faiz iade talep et.

domain-specific keywords to address this limitation. The keywords included terms such as “sonuç” (result), “ret” (rejection), and “kabul” (acceptance), which are central to understanding judicial rulings. When these keywords were applied, the algorithms prioritized sentences containing these terms, improving the accuracy of the extracted summaries by focusing on content directly related to judicial decisions. The combination of TF-IDF and TextRank offers significant

advantages over generic summarization methods. TF-IDF quantifies the importance of terms based on their frequency within a document relative to their occurrence across a collection of documents, making it practical for identifying representative sentences. TextRank, in contrast, employs a graph-based ranking model to capture relationships between sentences, highlighting the most relevant ones within a document. These methods are computationally efficient, unsupervised, and adaptable, making them particularly suited for processing legal documents’ structured and repetitive nature.

The summarization process achieved greater precision and relevance by integrating domain-specific keywords into these techniques, ensuring that critical legal terms were emphasized. This approach outperformed generic summarization methods by delivering summaries that accurately captured the essence of judicial decisions, which is essential in domains like law, where specific terminology carries significant weight.

1) TF-IDF METHOD

Term Frequency-Inverse Document Frequency (TF-IDF) [43] is a statistical measure used to evaluate the importance of words within a document relative to a collection of documents (corpus). TF-IDF highlights significant terms in a document, but these terms are not necessarily frequent across the entire dataset. The TF-IDF score for each term t in document d is calculated as follows:

$$\text{TF-IDF}(t, d, D) = \text{TF}(t, d) \times \text{IDF}(t, D) \quad (1)$$

where t is the term (word), d is a specific document in the corpus, D is the total number of documents in the corpus. The Term Frequency (TF) measures how often a term appears in a document:

$$\text{TF}(t, d) = \frac{\text{Number of occurrences of } t \text{ in } d}{\text{Total number of terms in } d} \quad (2)$$

The Inverse Document Frequency (IDF) reflects how unique or rare a term is across all documents in the corpus:

$$\text{IDF}(t, D) = \log \left(\frac{N}{1 + |\{d \in D : t \in d\}|} \right) + 1 \quad (3)$$

where N is the total number of documents in the corpus D , and $|\{d \in D : t \in d\}|$ is the number of documents in which the term t appears.

2) TEXTRANK METHOD

TextRank [44] is an unsupervised graph-based algorithm used for keyword extraction and text summarization by ranking sentences or words based on their importance in the text through the analysis of their relationships. The TextRank score $S(V_i)$ for each sentence (V_i) is calculated iteratively as follows:

$$S(V_i) = (1 - d) + d \times \sum_{v_j \in \text{In}(V_i)} \frac{S(V_j)}{|\text{Out}(V_j)|} \quad (4)$$

where $S(V_i)$ is the score of sentence V_i , d is the damping factor (usually set to 0.85), $\text{In}(V_i)$ is the set of sentences that link to V_i , $\text{Out}(V_j)$ is the number of outgoing links from sentence V_j . The similarity between pairs of sentences was calculated using cosine similarity, comparing word content overlap. Sentences were then represented as nodes in a graph, with edges weighted by their similarity scores.

3) EVALUATION OF EXTRACTIVE METHODS: TF-IDF VS. TEXTRANK

The performance of the extractive summarization methods, TF-IDF and TextRank, was evaluated using several content-based metrics: Cosine Similarity, Content Precision, Content Recall, and Content F1-Score. These metrics compared machine-generated summaries against reference summaries created by legal experts, which included essential elements such as the acceptance or rejection of lawsuits, evidence evaluation, and final decisions. TF-IDF relies on statistical calculations based on term frequency to identify key terms. While this approach effectively highlights frequently occurring terms, it can lead to overrepresenting standard, non-distinctive terms, resulting in repetitive content. This limitation may hinder its ability to capture the essence of legal documents, mainly when critical elements, such as final decisions, do not occur frequently enough. However, incorporating domain-specific keywords into the TF-IDF process improved its performance, enabling it to focus on terms central to judicial rulings. Despite this enhancement, TF-IDF's dependence on frequency metrics may limit its ability to understand the legal context comprehensively. TextRank, a graph-based unsupervised approach, ranks sentences based on their importance by analyzing relationships between sentences. Unlike TF-IDF, it does not rely solely on term frequency, reducing the risk of overemphasizing common words. This makes TextRank more effective at capturing sentence importance in a broader context. However, TextRank may face challenges in fully representing the legal context, particularly when identifying critical legal terms directly tied to final decisions. The evaluation results, summarized in Table 2, highlight the strengths and limitations of both methods:

TABLE 2. Performance comparison of TF-IDF and TextRank for legal document summarization.

Metrics	TF-IDF		TextRank	
	without keywords	with keywords	without keywords	with keywords
Cosine similarity	0.23	0.44	0.43	0.33
Content Precision	0.16	0.40	0.34	0.23
Content Recall	0.24	0.56	0.58	0.41
Content F1-Score	0.19	0.47	0.43	0.29

- Cosine Similarity:
 - TF-IDF with keywords achieved a moderate similarity score of 0.33, indicating its ability to align with reference summaries.
 - TextRank with keywords outperformed TF-IDF, achieving a higher similarity score of 0.44,

reflecting its effectiveness in preserving relationships between key terms and the overall summary.

- Without keywords, TF-IDF performed poorly (0.23), whereas TextRank maintained its performance (0.44).
- Content Precision:
 - TF-IDF with keywords achieved the highest precision (0.40), demonstrating its ability to select the most relevant content for summarization.
 - TextRank with keywords had a lower precision (0.23), as it captured broader contextual information but included more irrelevant or repetitive details.
 - Without keywords, both methods underperformed, with TF-IDF scoring 0.16 and TextRank scoring 0.34.
- Content Recall:
 - TF-IDF with keywords captured a higher proportion of relevant content (0.56) than TextRank with keywords (0.41), highlighting its strength in identifying essential elements, especially when guided by keywords.
 - Without keywords, TextRank outperformed TF-IDF (0.58 vs. 0.24), demonstrating its ability to capture a wider range of content when not constrained by term frequency.
- Content F1-Score:
 - The F1-Score, which balances precision and recall, was highest for TF-IDF with keywords (0.47), indicating its superior ability to balance capturing relevant content and minimizing irrelevant details.
 - TextRank with keywords achieved a lower F1-Score (0.29), reflecting its trade-off between strong recall and weaker precision.

The evaluation demonstrates that when combined with keywords, TF-IDF excels in precision and overall balance between precision and recall. It is well-suited for tasks requiring focused extraction of legal content. Conversely, TextRank's unsupervised graph-based approach is more effective at maintaining contextual relationships and capturing broader content, mainly when keyword guidance is unavailable. These insights underscore the complementary strengths of the two methods and suggest the potential for hybrid approaches to enhance summarization performance in the legal domain further.

4) ANALYSIS OF GENERATED SUMMARY

To evaluate the effectiveness of the generated summaries, we compared the outputs of TF-IDF and TextRank against expert-crafted reference summaries. Each reference summary included the following critical components:

- The acceptance or rejection of the lawsuit.
- Evaluation of evidence.
- The final decision, including any appeals or rulings.

Despite its limitations, TF-IDF strongly aligned with expert-crafted summaries, particularly when domain-specific keywords were incorporated. This method effectively retained key legal decisions, especially the final rulings. However, in the absence of keywords, TF-IDF often overrepresented standard, less relevant terms, negatively impacting the summary's quality and focus. Notably, TF-IDF without keywords failed to consistently capture critical sentences, such as those containing "sonuç" (the court's decision), essential for summarizing legal documents. In contrast, TF-IDF with keywords successfully highlighted this key sentence, underscoring the importance of keyword guidance in improving the method's performance.

TextRank demonstrated its strength in capturing broader contextual relationships between sentences, making it particularly effective for summarizing complex documents with diverse content. Interestingly, TextRank without keywords could identify critical sentences, such as those containing "sonuç," even without keyword guidance. This highlights its ability to capture essential legal details through its graph-based ranking approach. However, when introducing keywords, TextRank sometimes failed to highlight critical sentences like "sonuç." This limitation is likely due to the influence of keyword weighting, which shifted the algorithm's focus to other, less relevant sentences.

The choice of technique depends on the specific requirements of the summarization task:

- When precision and recall of critical legal elements (e.g., the court's decision) are prioritized, TF-IDF with keywords is the most effective. It consistently captures key details while minimizing irrelevant content.
- When broader contextual understanding and coverage of related content are more important, TextRank without keywords is preferable. It can identify critical sentences like "sonuç" even without keyword guidance, though at the cost of lower precision.

Neither technique can be deemed universally superior; their effectiveness depends on the context and objectives of the summarization task. A performance comparison of TF-IDF (with and without keywords) and TextRank (with and without keywords) for legal document summarization is shown in Figure 1. Future work should explore hybrid approaches that combine the precision of TF-IDF with the contextual fluency of TextRank. Additionally, enhancing both methods to consistently identify critical sentences, such as "sonuç," across all configurations will improve the overall quality of legal document summarization.

D. ABSTRACTIVE SUMMARIZATION

Following the generation of extractive summaries, the next step was abstractive summarization. This method creates new sentences to condense and coherently convey the meaning of source texts. This approach is beneficial in summarizing complex legal cases, often involving multiple related documents with overlapping or complementary details. Abstractive

summarization enables the synthesis of critical arguments, rulings, and legal reasoning into a concise summary. We utilized four advanced transformer-based models: LED-base-16384, Long-T5-tglobal-base, BART-large, and GPT-3.5 Turbo. Each model brings distinct strengths to address the challenges of summarizing long sequences, integrating information from multiple sources, and condensing complex legal arguments.

1) PROPOSED MODEL

- LED-base-16384 (Longformer Encoder-Decoder) [41] was selected for its ability to handle long sequences of up to 16,384 tokens, making it ideal for summarizing legal cases that span multiple documents. This model effectively integrates details from various documents, ensuring that critical information is not lost. It employs sparse attention, which reduces the computational complexity from $O(n^2)$ to $O(n)$, allowing LED to process extensive legal texts efficiently:

$$A_{\text{sparse}}(Q, K, V) = \text{sparse-softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (5)$$

where Q , K , and V are the query, key, and value matrices, and d_k is the dimension of the critical vectors. This architecture allows LED to summarize content from multiple documents while maintaining coherence and detail.

- Long-T5-tglobal-base (Text-to-Text Transfer Transformer) [46] supports sequences of up to 4096 tokens, making it suitable for summarizing moderately long collections of legal documents. Its text-to-text framework allows it to integrate information from multiple related documents into a unified summary. Long-T5 utilizes global attention, which helps the model focus on the most critical information from each document, ensuring that the final summary captures the core arguments and rulings.
- BART-large (Bidirectional Auto-Regressive Transformer) [47] processes sequences of up to 1024 tokens. While BART-large (Bidirectional Auto-Regressive Transformer) processes sequences of up to 1024 tokens, which is smaller than the limits of some other models such as LED-base-16384 or Long-T5-tglobal-base, it is highly effective for summarizing shorter collections of documents or documents that have already been condensed. BART-large's bidirectional encoder reads the full context of each input, and its auto-regressive decoder generates coherent and concise summaries. BART-large excels at producing readable and compact summaries, particularly for smaller or pre-condensed inputs.
- Generative Pre-trained Transformer 3.5 Turbo [48]: As a cutting-edge model, GPT-3.5 Turbo combines state-of-the-art natural language processing capabilities with an advanced understanding of contextual relationships.

REFERENCE SUMMARY:

[illegible]

TF-IDF WITHOUT KEYWORDS:

[illegible]

TEXTRANK WITHOUT KEYWORDS:

[illegible]

TF-IDF WITH KEYWORDS:

[illegible]

TEXTRANK WITH KEYWORDS:

[illegible]

FIGURE 1. Visual comparison of reference, TF-IDF, and TextRank summaries for legal document summarization.

Its ability to process and generate coherent summaries from complex and lengthy legal texts makes it a superior choice for abstractive summarization. The model is particularly effective in generating human-like, semantically rich summaries, outperforming earlier models in precision and contextual accuracy.

2) TRAINING PROCESS

Each model was fine-tuned to summarize multiple related legal documents into a single, concise, and informative summary that accurately reflected the legal decisions, arguments, and outcomes across the documents.

- **Input and Output Data**
 - **Input:** The input to the models consisted of the extractive summaries generated from multiple legal

documents for each case. These extractive summaries provided a concise version of the relevant sections of the legal texts, allowing the models to condense them further.

- **Output:** The output was a single abstractive summary combining the key points from multiple documents into a cohesive, human-readable format. The generated summaries were compared against human-written summaries to ensure accuracy and fidelity to the legal content.

- Fine-Tuning Procedure
 - Loss Function: The models were fine-tuned using the cross-entropy loss function. For each token y_t ,

the loss is computed as:

$$L = - \sum_{t=1}^T \log P(y_t | y_1, y_2, \dots, y_{t-1}, x) \quad (6)$$

x is the input (a set of extractive summaries from multiple documents), and y_t is the predicted token at time step t . The training goal was to minimize this loss, ensuring the generated summaries aligned closely with the human-generated reference summaries.

- Beam Search: Beam search was employed to generate the final summaries during inference. Beam search allows the model to consider multiple potential summaries at each step and select the most probable one based on cumulative likelihood.

3) ABSTRACTIVE SUMMARIZATION PROCESS

The abstractive summarization process involved the following steps:

- Input: The process began with extractive summaries from multiple legal documents. These input summaries captured the most relevant sentences from each document, serving as a foundation for further abstraction.
- Fine-Tuning: Abstractive models were fine-tuned to accurately reflect the legal context, rulings, and arguments across multiple documents. The fine-tuning process optimized the models to consolidate information, eliminate redundancy, and preserve essential points critical to legal analysis.
- Output: The final output was an abstractive summary that condensed vital information from multiple legal documents into a coherent and concise format. These summaries retained the accuracy and integrity of the original legal texts while presenting the information in a more accessible manner.

Future improvements will focus on incorporating domain-specific pretraining for transformer models, such as GPT-4 or LongT5, fine-tuned on Turkish legal corpora. This targeted pretraining addresses challenges posed by Turkish's agglutinative morphology and specialized legal terminology. Enhancing semantic consistency and coherence will improve the quality and relevance of the generated summaries, making them more suitable for legal applications.

E. EVALUATION METRICS

To evaluate the quality of the generated summaries, we used the ROUGE (Recall-Oriented Understudy for Gisting Evaluation) metric [49], which is widely used in text summarization tasks to assess how well a machine-generated summary captures the content of a reference summary. ROUGE measures the overlap between words or phrases in the generated and reference summaries, providing a quantitative similarity measure. Below are the primary ROUGE metrics used in this study, along with their mathematical definitions:

1) ROUGE-1 (UNIGRAM OVERLAP)

ROUGE-1 measures the unigrams (single words) overlap between the generated and reference summaries. It provides insight into how well individual words from the reference summary are captured in the generated summary. The formula for ROUGE-1 recall is:

$$\text{ROUGE-1 (Recall)} = \frac{\sum_{\text{unigram} \in \text{Ref}} \min(\text{Count}_{\text{Ref}}(\text{unigram}), \text{Count}_{\text{Gen}}(\text{unigram}))}{\sum_{\text{unigram} \in \text{Ref}} \text{Count}_{\text{Ref}}(\text{unigram})} \quad (7)$$

where $\text{Count}_{\text{Ref}}(\text{unigram})$ is the number of times a unigram appears in the reference summary. $\text{Count}_{\text{Gen}}(\text{unigram})$ is the number of times the same unigram appears in the generated summary. Similarly, precision and F1-score can also be calculated for ROUGE-1. The F1-score is often used as a harmonic mean of recall and precision.

2) ROUGE-2 (BIGRAM OVERLAP)

ROUGE-2 measures the overlap of bigrams (two consecutive words) between the generated and reference summaries. This metric reflects how well the model captures short phrases, contributing to summary coherence. The formula for ROUGE-2 recall is:

$$\text{ROUGE-2 (Recall)} = \frac{\sum_{\text{bigram} \in \text{Ref}} \min(\text{Count}_{\text{Ref}}(\text{bigram}), \text{Count}_{\text{Gen}}(\text{bigram}))}{\sum_{\text{bigram} \in \text{Ref}} \text{Count}_{\text{Ref}}(\text{bigram})} \quad (8)$$

where a bigram refers to two consecutive words, this score helps evaluate the preservation of keyword pairs from the reference summary in the generated summary.

3) ROUGE-L (LONGEST COMMON SUBSEQUENCE)

ROUGE-L measures the Longest Common Subsequence (LCS) between the generated and reference summaries, considering sentence-level structure and word order. This metric is more sensitive to fluency and coherence. The formula for ROUGE-L recall is:

$$\text{ROUGE-L (Recall)} = \frac{\text{LCS}(\text{Ref}, \text{Gen})}{\text{Length}(\text{Ref})} \quad (9)$$

where $\text{LCS}(\text{Ref}, \text{Gen})$ is the length of the longest common subsequence shared between the reference and generated summaries, $\text{Length}(\text{Ref})$ is the total number of tokens in the reference summary.

4) ROUGE-SUM (COMPOSITE SCORE)

ROUGE-Sum provides an aggregated measure by averaging the ROUGE-1, ROUGE-2, and ROUGE-L scores. This composite score gives a holistic view of the model's summarization performance:

$$\text{ROUGE-Sum} = \frac{\text{ROUGE-1} + \text{ROUGE-2} + \text{ROUGE-L}}{3} \quad (10)$$

TABLE 3. ROUGE metric scores for LED, Long-T5, BART-large, and GPT-3.5 turbo in legal document summarization before and after training.

Model Name	ROUGE-1	ROUGE-2	ROUGE-L	ROUGE-SUM
before training	LED	0.22	0.07	0.20
	Long-T5	0.25	0.08	0.22
	BART-large	0.20	0.05	0.18
	GPT-3.5 Turbo	0.28	0.10	0.25
after training	LED	0.45	0.25	0.33
	Long-T5	0.40	0.20	0.30
	BART-large	0.35	0.14	0.20
	GPT-3.5 Turbo	0.55	0.35	0.44

This metric simplifies the evaluation process by providing a single score that reflects the overall quality of the generated summary based on content overlap and structural retention. ROUGE metrics offer a comprehensive view of summarization quality:

- ROUGE-1 provides basic content coverage, ensuring that keywords are included in the summary.
- ROUGE-2 ensures that short, meaningful phrases are retained, contributing to better coherence.
- ROUGE-L goes beyond simple word matching by accounting for word sequence and structure, ensuring that the summary flows logically and maintains the structure of the original text.

These metrics, when combined, offer a holistic evaluation of how well the model-generated summaries align with human-written reference summaries. Combining these metrics is especially useful in legal documents, where content accuracy and structure are crucial. While ROUGE metrics provide valuable quantitative assessments, future evaluations will incorporate human judgment by legal professionals to assess the summaries' semantic accuracy, contextual appropriateness, and ability to preserve critical legal nuances. This combination of quantitative and qualitative metrics will provide a more comprehensive model performance evaluation.

III. RESULTS AND DISCUSSION

The effectiveness of the summarization framework is evaluated using several ROUGE metrics that assess different aspects of summary quality. These metrics provide complementary perspectives on how well the generated summaries capture the content and structure of reference summaries.

A. ROUGE METRICS PERFORMANCE

Table 3 presents the ROUGE metric scores for LED, LongT5, BART-large, and GPT-3.5 Turbo before and after fine-tuning. The results demonstrate that all models benefited from fine-tuning, showing improvements in content recall, phrase-level coherence, and structural alignment with human-written summaries, albeit to varying extents.

B. MODEL COMPARISON BASED ON ROUGE SCORE

The ROUGE metrics provide the following insights into the performance of each model:

- **LED: Strong Post-Training Gains, Second-Best Performance Overall**
 - Pre-training Performance: LED exhibited limited summarization capability, with a ROUGE-1 score of 0.22, ROUGE-2 score of 0.07, and ROUGE-L score of 0.20, indicating poor content retention and structural coherence.
 - Post-training Performance: LED showed substantial improvements after fine-tuning, achieving a ROUGE-1 score of 0.45 (104.5% increase), ROUGE-2 score of 0.25, and ROUGE-L score of 0.33. These results reflect better phrase-level understanding and structural consistency.
 - While LED performed well after fine-tuning, it was outperformed by GPT-3.5 Turbo across all metrics.
- **LongT5: Moderate Improvements, Slightly Below the LED**
 - Pre-training Performance: LongT5 achieved a ROUGE-1 score of 0.25, ROUGE-2 score of 0.08, and ROUGE-L score of 0.22. These scores positioned it slightly above LED in unigram recall but highlighted its phrase and sentence coherence limitations.
 - Post-training Performance: Fine-tuning improved LongT5 to ROUGE-1: 0.40, ROUGE-2: 0.20, and ROUGE-L: 0.30, demonstrating meaningful but more moderate gains than LED.
 - Despite notable improvements, LongT5's final performance was slightly inferior to LED, suggesting its potential for improvement in legal document summarization.
- **BART-large: Weakest Performance Despite Some Improvement**
 - Pre-training Performance: BART-large scored lowest, with ROUGE-1 at 0.20, ROUGE-2 at 0.05, and ROUGE-L at 0.18, reflecting weak content recall and structural coherence.
 - Post-training Performance: After fine-tuning, BART-large improved to ROUGE-1: 0.35, ROUGE-2: 0.14, and ROUGE-L: 0.20. However, it remained the weakest model overall.
 - Although BART-large benefited from fine-tuning, its performance was suboptimal compared to the other models, indicating it may not be suitable for legal document summarization tasks.
- **GPT-3.5 Turbo: Most Significant Improvement and Best Overall Performance**
 - Pre-training Performance: GPT-3.5 Turbo performed moderately well before training, with a ROUGE-1 score of 0.28, ROUGE-2 score of 0.10, and ROUGE-L score of 0.25.

TABLE 4. Performance improvements for various models across ROUGE metrics.

Model	ROUGE-1 Improvement	ROUGE-2 Improvement	ROUGE-L Improvement
LED	+0.23 (104.5%)	+0.18 (257%)	+0.13 (65.0%)
Long-T5	+0.15 (60.0%)	+0.12 (150%)	+0.08 (36.4%)
BART-large	+0.15 (75.0%)	+0.09 (180%)	+0.02 (11.1%)
GPT-3.5 Turbo	+0.27 (96.4%)	+0.25 (250%)	+0.17 (68.0%)

- Post-training Performance: GPT-3.5 Turbo achieved the most substantial improvements after fine-tuning, with a ROUGE-1 score of 0.55, ROUGE-2 score of 0.35, and ROUGE-L score of 0.42. These results reflect superior content retention, phrase-level coherence, and structural consistency.
- GPT-3.5 Turbo demonstrated the highest performance post-training, making it the most effective model for legal document summarization.

The improvements in ROUGE scores highlight the critical role of fine-tuning in enhancing summarization performance. Table 4 presents the absolute improvements for each model, underscoring their relative strengths and weaknesses. While GPT-3.5 Turbo outperformed all other models across all metrics, LED, and LongT5 showed competitive results after training. In contrast, despite some improvement, BART-large remained the least effective for summarizing legal documents.

Table 4 summarizes the performance improvements of various models across ROUGE-1, ROUGE-2, and ROUGE-L metrics, with both absolute and percentage gains calculated using the scores from Table 3. The percentage improvement is determined using the formula:

$$\text{Percentage Improvement} = \frac{\text{After Training Score} - \text{Before Training Score}}{\text{Before Training Score}} \times 100\% \quad (11)$$

For example, the ROUGE-1 improvement for LED in Table 4 was derived as follows based on the scores in Table 3:

$$\frac{0.45 - 0.22}{0.22} \times 100\% = 104.5\% \quad (12)$$

The results in Table 4 highlight the performance improvements achieved by each model after fine-tuning, with specific insights into their ability to enhance summarization metrics:

- LED: Demonstrated significant gains, particularly in ROUGE-2 (+0.18, 257%), reflecting improved phrase-based summarization performance.
- Long-T5: Showed moderate improvements, including ROUGE-1 (+0.15, 60.0%) and ROUGE-2 (+0.12, 150%). However, its structural coherence remained a challenge, as evidenced by a smaller ROUGE-L improvement (+0.08, 36.4%).
- BART-large: Exhibited the weakest performance, with minimal gains in ROUGE-L (+0.02, 11.1%), indicating persistent structural alignment and content retention difficulties.

TABLE 5. Comparison of document reduction percentages across TF-IDF, LED, Long-T5, BART, and GPT-3.5 Turbo summarization methods.

	Extractive Summary	LED	Long-T5	BART-large	GPT-3.5 Turbo
Reduction Percentage	~ 67.65%	~ 86.55%	~ 92.76%	~ 85.38%	~ 90.14%

- GPT-3.5 Turbo: Achieved the highest improvements across all metrics, with notable gains in ROUGE-1 (+0.27, 96.4%) and ROUGE-2 (+0.25, 250%). These results underscore its significant advancements in basic content coverage and phrase-level coherence.

These findings emphasize the importance of fine-tuning models for domain-specific tasks like legal document summarization. While GPT-3.5 Turbo emerged as the most effective model, LED and Long-T5 also demonstrated promising results. Conversely, BART-large's weaker performance suggests further optimization for complex legal text tasks.

C. APPROXIMATE WORD REDUCTION

Another critical aspect of summarization is condensing original content while preserving essential information. Table 5 outlines the approximate word reduction achieved by each model for extractive and abstractive summarization models.

- LED: Reduced the input content, derived from the output of extractive summarization, by approximately 86.55%, effectively retaining the most important legal information.
- Long-T5: Achieved the highest reduction at 92.76% based on the input derived from the output of extractive summarization. However, this extreme reduction may have contributed to its slightly lower ROUGE scores, as excessive condensation can compromise content retention.
- BART-large: Reduced the input content, derived from the output of extractive summarization, by 85.38%. Despite this reduction, its lower ROUGE scores indicate struggles in preserving critical legal details.
- GPT-3.5 Turbo: Reduced the input content, derived from the output of extractive summarization, by 76.14%, the least aggressive reduction among the models. This more conservative approach allowed it to balance summarization and retaining critical details, contributing to its competitive performance.

These results suggest that while aggressive content reduction can enhance brevity, it may also lead to a loss of critical information. GPT-3.5 Turbo's balanced approach highlights the importance of prioritizing content retention in tasks requiring high precision, such as legal document summarization.

D. QUALITATIVE COMPARISON OF SUMMARY

In addition to quantitative metrics, a qualitative assessment was conducted to evaluate the summaries generated by LED, Long-T5, BART-large, and GPT-3.5 Turbo for a specific legal case. This assessment considered differences in summary

length, content coverage, and coherence. Table 6 provides detailed comparisons of the models' outputs in terms of:

- **Content Retention:** The ability to include key legal details, such as rulings, evidence evaluations, and final decisions.
- **Accuracy:** The correctness of the information presented in the summaries.
- **Logical Flow:** The coherence and readability of the summaries.

The qualitative analysis revealed that GPT-3.5 Turbo consistently produced summaries with superior content retention and logical flow, making it the most effective model for generating accurate and coherent legal summaries. LED and Long-T5 demonstrated reasonable performance but occasionally lacked structural coherence. While improved post-training, BART-large struggled to generate concise and accurate summaries, reinforcing its weaker performance observed in the quantitative evaluation.

E. LIMITATIONS OF ROUGE METRICS IN LEGAL SUMMARIZATION

ROUGE metrics offer a useful framework for evaluating summarization models, but they fall short of capturing the full quality of summaries, particularly in specialized domains such as legal document summarization. While ROUGE measures word overlap, legal summaries require more than just lexical similarity—they must preserve the original text's legal meaning, nuances, and structural integrity. For example, despite LED achieving high ROUGE scores by producing detailed and comprehensive summaries, other models like Long-T5, BART-large, and GPT-3.5 Turbo also generated valuable summaries, each with distinct characteristics. GPT-3.5 Turbo, for instance, demonstrated a unique ability to generate fluent, human-like summaries that balanced conciseness with comprehensiveness. However, while it achieved the highest ROUGE scores overall, its occasional paraphrasing of legal terms highlighted the metric's limitations in capturing domain-specific precision. This example underscores that high ROUGE scores do not always reflect a model's capacity to maintain the semantic and legal fidelity critical in legal contexts. Therefore, additional qualitative evaluation methods, such as human review by legal professionals, are essential to complement ROUGE metrics for a holistic assessment of summarization performance. Below is a comparison of their outputs for the same legal case:

- **LED Summary:** LED maintained a detailed narrative of the legal proceedings, including case rulings and appeals. Its summary accurately reflected the entirety of the case but was relatively verbose.
- **Long-T5 Summary:** Long-T5 produced a concise summary, focusing on key decisions such as the nullification of the contract and the judgment against the defendant. While it missed some procedural details, its structure and readability were strong.

- **BART-large Summary:** BART-large prioritized procedural aspects, detailing the appeals and court processes. It captured essential rulings but occasionally added optional procedural information, which may detract from the focus on core legal decisions.
- **GPT-3.5 Turbo Summary:** GPT-3.5 Turbo provided a concise and well-organized summary, effectively covering critical rulings and procedural details. However, despite its high ROUGE scores, it occasionally paraphrased legal terminology or omitted subtle nuances vital to the legal context.

This highlights a critical limitation of ROUGE metrics: high scores do not always correspond to a model's ability to preserve the semantic and legal integrity necessary for legal summarization.

F. INTERPRETATION OF MODEL PERFORMANCE

While GPT-3.5 Turbo achieved the highest ROUGE scores across all metrics, each model demonstrated distinct strengths depending on the summarization context and style:

- **LED:** Best suited for detailed and comprehensive summaries, LED captures procedural and legal details. However, its verbosity may include extraneous information that is not always essential.
- **Long-T5:** Ideal for scenarios requiring brevity, Long-T5 provides concise and structured summaries. Although it sacrifices some detailed procedural information, it ensures a clear and focused output, making it suitable for contexts where readability and succinctness are prioritized.
- **BART-large:** Designed for shorter summaries, BART-large effectively captures essential legal rulings while adding procedural context. However, this additional detail can sometimes lower ROUGE scores despite its contextual accuracy.
- **GPT-3.5 Turbo:** Combining conciseness with comprehensiveness, GPT-3.5 Turbo balances procedural details and key legal rulings. Although its high ROUGE scores reflect strong overall performance, it occasionally misses subtle legal nuances that are critical in specific cases.

In the legal domain, the quality of a summary extends beyond word overlap. Legal precision, contextual understanding, and the preservation of nuances are paramount. ROUGE scores alone cannot fully evaluate the effectiveness of a summarization model, particularly when variations in summary length, complexity, and procedural emphasis arise. To ensure high-quality outputs, it is essential to supplement ROUGE evaluations with manual assessments. Human reviewers can determine whether a summary accurately captures the legal logic, context, and outcomes, ensuring that the model's outputs align with the specific demands of legal documentation.

TABLE 6. Comparison of summaries generated by LED, Long-T5, BART-large, and GPT-3.5 turbo models.

Model Name	Generated Summary	Translated Summary (English)	Comments/Analysis
LED	İki davada da davacılar, devremülk satış vaadi sözleşmesinin feshi ve davacının borçlu olmadığının tespiti talep etmiştir. İlk davada, davacı devremülk teslim edilmediğini belirterek ödediği yasal faiziyle birlikte davalıdan tahsil edilmesini talep etmiştir. Mahkeme, davanın kabulüne ve devremülk satış vaadi sözleşmesinin feshi ve davacının borçlu olmadığının tespitine karar vermiş, ancak davalı temyiz etmiştir. İkinci davada ise, davacı devremülk satış vaadi sözleşmesinin feshi ve davacının borçlu olmadığının tespitine karar vermiştir. Mahkeme, davalıya borçlu olmadığının tespitine ve dava tarihinden itibaren işleyecek avans faiziyle davalıdan tahsiline karar vermiş, ancak temyiz sonucunda hüküm davalı lehine bozulmuştur.	In both cases, the plaintiffs requested the termination of the timeshare sales promise agreement and the determination that the plaintiff is not indebted. In the first lawsuit, the plaintiff stated that the timeshare was not delivered and demanded the collection from the defendant together with the legal interest paid. The court decided to accept the lawsuit and to terminate the timeshare sales promise agreement and to determine that the plaintiff is not indebted, but the defendant appealed. In the second case, the plaintiff decided to terminate the timeshare sales contract and to determine that the plaintiff is not indebted. The court decided to determine that the plaintiff was not indebted to the defendant and to collect from the defendant with the advance interest to be accrued from the date of the lawsuit, but the judgement was reversed in favour of the defendant on appeal.	The LED model provides a detailed and comprehensive summary, capturing all procedural details, including appeals and their outcomes. However, the summary is verbose and repetitive, with redundant phrases that make it less concise and harder to read. This suggests the model is heavily influenced by the extractive input without sufficient abstraction.
Long-T5	Taraflar arasındaki devremülk satış vaadi sözleşmesinin feshi ve davacının davalıya borçlu olmadığının tespitine karar verilmiştir. Davacının davalıya borçlu olmadığının tespitine ve dava tarihinden itibaren avans faiziyle birlikte davalıdan tahsiline karar verilmiştir. Mahkeme davanın kabulüne karar vermiş olup, devremülk satış vaadi sözleşmesinin feshi ve davacının davalıya borçlu olmadığının tespitine hükmedilmiştir.	It was decided to terminate the timeshare sales contract between the parties and to determine that the plaintiff was not indebted to the defendant. It was decided to determine that the plaintiff was not indebted to the defendant and to collect from the defendant with advance interest from the date of the lawsuit. The court decided to accept the lawsuit and ruled for the termination of the timeshare sales contract and the determination that the plaintiff is not indebted to the defendant.	Long-T5 generates a concise and focused summary, effectively capturing key decision points. However, it omits some procedural details, such as the defendant's appeal and its reversal, which reduces contextual completeness. The model performs well in abstracting the extracted input into a shorter and more structured output but sacrifices some nuanced elements.
BART-large	İki davada da davacılar, devremülk sözleşmelerinin iptali ve ödenen bedelin yasal faiziyle birlikte geri alınması talebiyle açılan davalarda, mahkemeler davanın kabulüne karar vermiştir. Davalı taraf temyiz etmiş, ancak mahkeme kararları usul hatası nedeniyle bozma sebebi sayılmıştır. Temyiz incelemesi sonucunda, dava tarihinden itibaren işleyecek avans faizinin davalıdan tahsil edilmesine hükmedilmiş ve hükmün davalıya borçlu olmadığının tespiti yönünden itirazların reddedilmesi gerektiği belirtilerek, davaların bozulmasına ve dosyanın yetersiz olduğu tespit edilmektedir. Her iki kararda da mahkemenin usul yasasına aykırı bulunarak, düzeltme yolu açma hakkına sahip olmaksızın vurgulanmış ve peşin alınan harcı iadesini talep ettiği için dosyalar ilgili yeterince yeterli mahkemelere yönlendirilmesinde görülmüştür.	In both cases, the plaintiffs filed lawsuits for the cancellation of the timeshare contracts and the recovery of the price paid with legal interest, and the courts decided to accept the lawsuits. The defendant appealed, but the court decisions were considered as a reason for reversal due to procedural errors. As a result of the appellate review, it was ruled that the advance interest to be processed as of the date of the lawsuit should be collected from the defendant and the objections should be rejected in terms of determining that the judgement is not indebted to the defendant, and the cases are reversed and the file is determined to be insufficient. In both decisions, it has been emphasised that the court is found to be contrary to the procedural law, without having the right to open a correction path, and the files have been seen in the referral of the files to the relevant sufficiently sufficient courts, as it has requested the return of the advance fee.	The BART-large model attempts to include broader procedural details, such as appeals and procedural errors, but its output lacks coherence. Long and cluttered sentences obscure the core legal rulings, and irrelevant details distract from the main points. The model struggles to abstract the extractive input effectively, resulting in summaries that are less focused and harder to interpret.
GPT-3.5 Turbo	Her iki belgede de, davacı devremülk sözleşmesinin iptali ve ödenen bedelin yasal faiziyle birlikte iadesini talep etmiştir. İlk belgede, davacı, davalı şirketin proje aşamasında avantajlı koşullarla termal kaplıca tesisinden hisse satın aldığını, ancak hisselerin devri ve devremülkün teslim edilmediğini belirterek ödemelerin iadesini talep etmiştir. İkinci belgede ise, mahkeme davanın kabulüne karar vermiş, devremülk satış vaadi sözleşmesinin feshi ve davacının davalıya borçlu olmadığının tespiti yönünde hüküm kurulmuştur. Her iki durumda da, davacının talepleri kabul edilmiş ve ödemelerin iadesi kararlaştırılmıştır.	In both documents, the plaintiff requested the cancellation of the timeshare contract and the refund of the amount paid together with legal interest. In the first document, the plaintiff stated that the defendant company purchased shares in the thermal spa facility with advantageous conditions during the project phase, but the shares were not transferred and the timeshare was not delivered, and requested the return of the payments. In the second document, the court decided to accept the lawsuit and ruled for the termination of the timeshare sales promise agreement and the determination that the plaintiff is not indebted to the defendant. In both cases, the demands of the plaintiff were accepted and the return of the payments was agreed.	GPT-3.5 Turbo generates fluent, well-structured summaries that balance readability and comprehensive coverage of key legal rulings and procedural details. It abstracts the extractive input effectively while maintaining coherence and human-like language. However, occasional paraphrasing of legal terms may dilute technical accuracy in some cases. Overall, it delivers the best performance by combining conciseness and comprehensiveness.

G. CHALLENGES IN MULTI-DOCUMENT SUMMARIZATION

This project presented several critical challenges, ranging from data collection and preprocessing to model selection and optimization. Below is an analysis of the primary issues encountered and the strategies implemented to address them.

1) COLLECTING AND CLEANING THE DOCUMENTS

A significant challenge involved collecting and preparing a dataset suitable for summarization. Legal documents often contain extensive metadata, procedural details, and extraneous information irrelevant to the core legal arguments. Ensuring the retention of only critical content—such as legal

decisions and arguments—required a robust cleaning process. A custom pipeline using the Zemberek library for Turkish tokenization was developed to address this. This pipeline effectively removed irrelevant metadata (e.g., dates, case numbers, procedural details) while preserving essential legal content. Text normalization techniques were also applied to ensure consistency, resulting in a clean and focused dataset tailored to the summarization models.

2) LANGUAGE AND LEGAL TERMINOLOGY

The Turkish language's complex grammatical structure and domain-specific legal terminology posed another significant challenge. The models needed to accurately understand and summarize terms commonly used in legal contexts, such as “Sonuç” (Result), “Ret” (Rejection), and “Kabul” (Acceptance). To address this, LED, Long-T5, BART-large, and GPT-3.5 Turbo were selected for their capabilities in handling specialized domains, including legal text. However, additional optimization steps were required. Custom preprocessing pipelines and legal-specific tokenization methods were applied to ensure these models could process legal terms accurately. By focusing the training data on a legal lexicon, the models' performance in summarizing legal arguments and decisions was significantly improved.

3) MANAGING LONG INPUT SEQUENCE

Legal documents are often lengthy and exceed most transformer models' token limits. This challenge became even more pronounced in multi-document summarization tasks, where multiple documents must be processed collectively. Model Capacities:

- LED and Long-T5: Both models can handle up to 2,048 tokens by default, with extended capacities of 16,384 tokens for LED and 4,096 tokens for Long-T5.
- BART-large: BART-large is more limited, with a maximum capacity of 1,024 tokens.
- GPT-3.5 Turbo: This model provides a token limit of 4,096 tokens, making it comparable to Long-T5 in input capacity.

To manage these constraints, preprocessing strategies were employed to prioritize relevant content and truncate less critical sections. This approach ensured that the most important legal arguments and procedural details were preserved within the token limits. However, input length limitations remained a persistent issue, particularly for complex multi-document cases, where critical details were occasionally omitted.

4) COMPUTATIONAL CHALLENGES: OUT OF MEMORY AND TRAINING EFFICIENCY

Training models on large datasets with lengthy input sequences posed computational hurdles, including GPU memory constraints and out-of-CUDA errors. To mitigate these issues:

- Gradient checkpointing and mixed precision training were implemented to reduce memory usage.

- Batch sizes and sequence lengths were optimized to balance computational efficiency with model performance.

These strategies allowed for the successful training of models like LED, which can process long documents without exceeding hardware limitations.

5) MANAGING TOKEN LIMITS AND ENSURING CONTENT COMPLETENESS

Managing token limits for lengthy legal documents proved challenging. When token capacities were insufficient, the resulting summaries were often incomplete, missing critical legal details. Strategies such as sequence truncation based on relevance were employed to address this, but further optimization is needed to ensure content completeness in future implementations.

6) ADDRESSING CHALLENGES IN LEGAL SUMMARIZATION

Token limitations in transformer models frequently lead to incomplete summaries of lengthy legal documents. Future work will explore solutions such as hierarchical summarization techniques and sequence truncation based on legal relevance. Additionally, the problem of hallucination in abstractive summaries can be mitigated by integrating factuality constraints and reinforcement learning techniques, enhancing the reliability of generated content.

7) LIMITATIONS IN DATASET

Despite the successful implementation of the hybrid framework, several limitations were identified:

- The dataset, while comprehensive for Turkish civil cases, does not cover other domains, such as criminal or constitutional law.
- Transformer-based models occasionally fail to preserve critical legal details in abstractive summarization.
- The preprocessing pipeline, though effective, could be further refined to capture the nuances of Turkish legal terminology better.

H. LIMITATIONS OF OUR HYBRID MODEL

While the proposed hybrid model demonstrates significant improvements in summarizing Turkish legal documents by combining extractive and abstractive methods, certain limitations merit further discussion:

- Dependence on Extractive Phase:
 - The quality of the hybrid model heavily depends on the extractive phase (TF-IDF or TextRank). If the extractive summarization fails to capture critical details or introduces irrelevant sentences, these issues propagate into the abstractive phase, limiting the overall effectiveness of the hybrid approach.
 - For example, TF-IDF may prioritize frequent terms while omitting legally significant but less frequent terms, and TextRank may focus on sentence relationships but fail to highlight key legal rulings.

- **Loss of Context During Abstraction:** The abstractive phase may rephrase or condense the extracted content in ways that lose critical nuances of legal language. This is particularly problematic in legal contexts where exact wording and technical accuracy are paramount.
- **Handling of Lengthy Documents:** Despite efforts to preprocess and prioritize content, managing long legal texts remains challenging. Extractive methods may truncate important sections, and abstractive models can only process a limited number of tokens, which may result in incomplete summaries.
- **Redundancy and Coherence:** While the hybrid model addresses the coherence issues typical in extractive summarization, redundancies from the extractive phase sometimes persist in the final output, leading to verbose or repetitive summaries.

I. FUTURE RESEARCH DIRECTION

To address these limitations, future work should explore the following:

- **Improved Integration of Extractive and Abstractive Methods:**
 - Develop dynamic hybrid models that iteratively refine extractive outputs before passing them to the abstractive phase. For instance, employing NER during extraction could ensure that critical legal terms and rulings are consistently included.
 - Introduce reinforcement learning-based strategies where the abstractive model provides feedback to improve the extractive phase.
- **Domain-Specific Pretraining:**
 - Fine-tune transformer models, such as GPT-4 or Long-T5, on a large corpus of Turkish legal texts to improve their understanding of the agglutinative morphology, legal terminology, and structural conventions unique to Turkish law.
 - This targeted pretraining would enhance extractive and abstractive phases by ensuring the models are better equipped to handle legal-specific nuances.
- **Hierarchical Summarization Techniques:**
 - Implement hierarchical models that summarize long legal texts in multiple stages. For instance:
 - * Stage 1: Extract summaries for individual sections or documents.
 - * Stage 2: Combine and refine these section summaries into a cohesive multi-document summary.
 - Such approaches would help manage token limits and ensure critical details from lengthy documents are not omitted.
- **Factual Consistency and Redundancy Reduction:**
 - Incorporate factual consistency checks to ensure that abstractive summaries do not introduce hallucinations or misinterpretations.

- Explore redundancy reduction algorithms during the extractive phase to minimize repetitive content and enhance coherence.
- **Expansion to Multilingual and Cross-Border Legal Cases:**
 - Extend the hybrid model to summarize legal documents in other underrepresented languages or cross-border cases, leveraging multilingual transformer models like mBART or XLM-Roberta.
 - Address challenges related to aligning legal terminology across languages to facilitate more effective summarization in international contexts.
- **Human-in-the-Loop Evaluation:** Engage legal professionals in the evaluation loop to provide feedback on generated summaries' legal accuracy, coherence, and relevance. This feedback can guide fine-tuning efforts and ensure the model aligns with the practical needs of the legal domain.

While the hybrid model represents a significant advancement in summarizing Turkish legal documents, its limitations underscore the need for continued refinement. Future research can further optimize the hybrid approach by integrating extractive and abstractive methods, leveraging domain-specific pretraining, and adopting innovative techniques such as hierarchical summarization and reinforcement learning. These advancements will improve the accuracy and coherence of legal summaries and extend the model's applicability to a broader range of legal contexts.

IV. CONCLUSION

This study presents a novel hybrid framework for multi-document summarization of Turkish legal texts, addressing a significant gap in legal NLP research. By integrating extractive techniques such as TF-IDF and TextRank with advanced transformer-based abstractive models—including LED, Long-T5, BART-large, and GPT-3.5 Turbo—we developed an effective methodology for generating concise and coherent legal summaries. Experimental results highlight the advantages of this approach, with GPT-3.5 Turbo achieving the highest ROUGE scores (ROUGE-1: 55%, ROUGE-2: 35%, ROUGE-L: 42%, and ROUGE-Sum: 44%), demonstrating its superior ability to produce fluent and accurate summaries. TF-IDF with keywords and TextRank without keywords excelled in precision and recall, respectively, ensuring a balance between factual accuracy and contextual fluency. Key contributions of this research include the following:

- **Dataset Creation:** The introduction of a curated dataset of 2,000 Turkish legal documents specifically designed for summarization tasks, filling a critical resource gap in Turkish legal NLP.
- **Hybrid Summarization Approach:** The development of a hybrid framework combining extractive content selection with abstractive generation, effectively addressing

the linguistic and structural challenges of Turkish legal texts.

- Transformer-Based Evaluation: A systematic evaluation of transformer-based models in the legal domain, with GPT-3.5 Turbo demonstrating state-of-the-art performance.
- Addressing Turkish Linguistic Challenges: The implementation of specialized preprocessing techniques tailored to the agglutinative nature of Turkish, setting a foundation for future research in low-resource legal languages.

Future work will focus on the real-world deployment of these models in legal environments, such as law firms and court systems, to assess practical challenges, usability, and impact on workflows. Enhancing hybrid techniques by integrating advanced extractive methods to guide abstractive generation will be explored to improve factual consistency and fluency further. Additionally, domain-specific pretraining of transformer models and expansion to larger, more diverse datasets, including criminal, administrative, and constitutional law, will be pursued to enhance model robustness across legal domains. Further advancements include the development of intelligent legal tools, such as question-answering systems based on judicial texts, and the integration of metadata extraction techniques for automated identification of court names, case numbers, legal precedents, and key arguments. Ethical considerations, including fairness, bias mitigation, and data privacy, will be prioritized to ensure professional and legal standards compliance. This research lays the groundwork for scalable, real-world applications that enhance legal workflows, improve access to justice, and support decision-making within Turkey's legal system by addressing these challenges.

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document summarization, focusing on hybrid transformer-based models, multi-document summarization, machine learning, and generative models.

MAHA AHMED ABDULLAH ALBAYATI

received the B.S. degree in computer engineering from Mosul University, Mosul, Iraq, in 2009, and the M.S. degree in computer engineering from Altinbas University, Istanbul, Türkiye, in 2021. She is currently pursuing the Ph.D. degree in computer engineering with Karabük University, Karabük, Türkiye. Her research interests include exploring the intersection of deep learning and natural language processing for legal



OGUZ FINDIK

received the Graduate degree from the Faculty of Engineering and Architecture, Department of Computer Engineering, Selçuk University, in 2000, and the master's and Ph.D. degrees in computer engineering from the Institute of Science, Selçuk University. Since 2020, he has been a Professor with the Department of Computer Engineering, Karabük University. His research interests include natural language processing, digital watermarking, artificial intelligence, machine learning, and swarm optimization algorithms. He has also led pioneering work in web programming, mobile programming, and game development using the unity game engine. In addition, he continues to work on projects related to artificial intelligence, image processing, and natural language processing. Over the years, he has contributed as a Project Lead, an Academic Advisor, and a Researcher in projects supported by TÜBİTAK BAP, and various development agencies.

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